

Phase field modeling of crack propagation in three-dimensional quasi-brittle materials under thermal shock

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ABSTRACT

Quasi-brittle materials create intricate crack patterns on their surface and internal three-dimensional crack networks when exposed to thermal shock conditions. Their cracking behavior is notably impacted by their quenching temperature and size. In this paper, a coupled three-dimensional phase field-cohesive zone model that accounts for thermoelastic fracture is developed to reproduce the quenching and cracking experiments of ceramic balls. The accuracy of the phase field model is demonstrated by simulating the phase field of a single-element cooling process and comparing it with the analytical solution. The quench fracture process of the 3D ceramic ball is then simulated. The phase field simulations show that the surface crack morphology during the quenching process is somewhat random. Still, the crack distribution density is determined by the quenching temperature difference and ceramic ball size. Furthermore, a threshold size is present during the quenching process. When the ceramic ball's radius falls below this threshold, the surface crack pattern no longer appears during quenching. Consequently, the phase-field-cohesive zone model represents an effective means of predicting three-dimensional fracture processes in quasi-brittle materials subjected to thermodynamically coupled loading.

1. Introduction

Under thermal shock, brittle and quasi-brittle solids often develop complex surface crack patterns and form complex three-dimensional crack networks within them. Examples include hot ceramic tiles in cold water [1–3], thermal shock of brittle and quasi-brittle solids [4–7], thermal shock of granite, limestone, sandstone and deep shale in various cooling media [8–10], and quenching of ceramic balls [11]. During these processes, thermal stresses cause cracks to initiate, propagate, coalesce and branch rapidly (typically within tens of milliseconds). The study of the mechanisms of crack formation during these processes has important implications for the design and application of thermally protected ceramic structures, and for the understanding and improvement of conventional ceramic manufacturing techniques.

Based on the principle of force-thermal equivalent energy density, Li et al. [12] established a new temperature-dependent failure criterion for the thermal impact fracture of ceramic materials. Additionally, it was proven that the tensile strain energy generated by thermal shock is the dominant mechanism for thermal shock-induced fracture. Shao et al. [11] conducted thermal shock experiments

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on ceramic balls. In this experiment, each size group was heated to a preset temperature. Afterward, the heated specimens were dropped in a free-fall manner into a thermostat. The specimens were impregnated with blue dye after drying to observe the cracks formed during the thermal shock. They experimentally demonstrated that the ceramic balls do not produce any cracks during thermal shock when their radius is less than a certain critical value. However, due to the complexity of the thermal shock fracture process under three-dimensional conditions, to our knowledge, no numerical studies are reproducing the above-mentioned three-dimensional crack extension process under quenching of ceramic balls and the size-dependent properties found in experiments.

The above experimental procedure demonstrates an interesting thermoelastic fracture phenomenon. The process is a typical problem of fracture of material under thermally coupled loading. Investigation by numerical simulation is an effective tool. Previous numerical simulation studies have been useful in exploring several related problems. Typical numerical methods include the boundary element method (BEM) [13–16], the extended finite element method (XFEM) [17–21], damage mechanics-based models [22–25], and the phase field models [26–31]. For example, Yan et al. [32] developed an extended finite element method model considering the effect of thermal coupling to study the fracture process of turbine blades, but it was limited to the extension study of individual cracks. Sun et al. [33] proposed a method that coupled peridynamic (PD) and finite element method (FEM) to study thermal cracking in brittle solids. Yan et al. [34] conducted a simulation of crack initiation and propagation in ceramic specimens during thermal shock using a finite element and discrete element coupling model. In addition, Yuan et al. [35] utilized the extended finite element method (XFEM) to estimate the initiation and propagation of cracks under thermal shock in a previously established microstructure model. As a result, the microscale failure mode is evident, and the influence of interface strength on the failure mode is examined. Li et al. [28] developed a phase field model under thermal coupling to study the quenching process of ceramic rods. Our previous work [36] developed the PF-CZM model to analyze the intricate crack pattern on the ceramic bottle surface during quenching, forming a solid foundation for the work presented in this manuscript.

However, no studies related to simulation have been conducted to precisely replicate the experimental phenomenon and present the procedure in detail. The primary challenges are listed as follows: (i) The fracture process is intricate, as it is a dynamic process under the influence of thermodynamic coupling loads. (ii) It is a three-dimensional asymmetric process that cannot be simplified to two-dimensional conditions. (iii) The crack pattern is multifaceted, featuring multiple three-dimensional spatial cracks that intersect, branch, and merge. In this paper, we have developed a phase field model for quasi-brittle materials that takes into account thermoelastic fracture and reproduced the ceramic quenching experiments of Shao et al. [11]. In addition, the influence of temperature and size of ceramic ball on the surface crack pattern formation during the quenching process of ceramic ball is studied. It is found that there is a size effect during quenching of ceramic balls.

2. The phase-field regularized cohesive zone model (PF-CZM)

In this section, the basic elements of current modeling frameworks that combine thermoelastic phase-field methods with cohesive region models are summarized. Particularly, the method is derived based on the theoretical achievements of Wu and Nguyen [37] as a reference, which makes a contribution to the combination of the phase field method for overall fracture and the consistent interface formulation relying on the classical cohesive zone model.

2.1. Variational formulation of thermo-elastic brittle fracture

The study examines the deformation and fracturing of a thermo-elastic solid in an Euclidean dimension space during the period $[0, t_f]$. Fig. 1 illustrates the presence or potential existence of a network of discrete cracks Γ_c within the solution domain. The solid is contained within the domain $\Omega \in R^d$, a three-dimensional sphere defined by the boundary $\partial\Omega$ with an external normal unit vector denoted by $\mathbf{n}$. Every point in the material located above is assigned a special position vector $\mathbf{x}$. The displacement field is represented by $\mathbf{u}(\mathbf{x}, t)$ while the temperature field is denoted as $T(\mathbf{x}, t)$. The assumption is that boundary conditions lead to the material being affected.

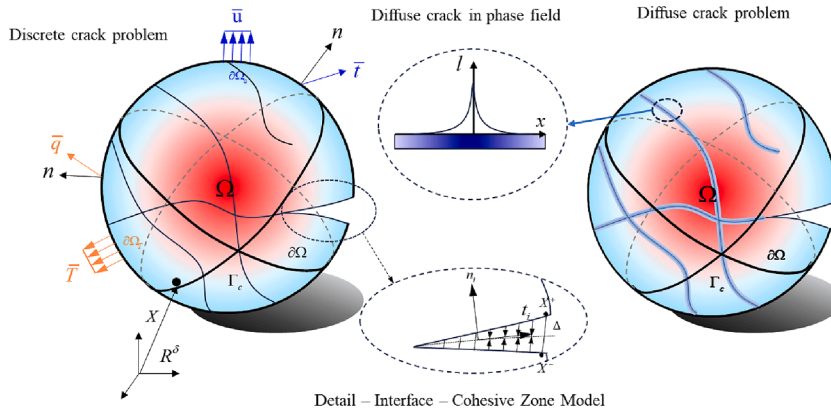


Fig. 1. Schematic diagram of any object with discontinuity in the domain under thermal shock load.

These boundary conditions are characterized by a specified displacement ($\bar{\mathbf{u}}$ on $\partial\Omega_u$) and a specified traction ($\bar{\mathbf{t}}$ on $\partial\Omega_t$). Additionally, the material is influenced by the specified temperature ($\bar{T}$ on $\partial\Omega_T$) and the specified heat flux ($\bar{\mathbf{q}}$ on $\partial\Omega_q$). These boundary conditions are governed by Neumann and Dirichlet conditions. In other words, the region in which they exist must satisfy the following relational equation:

$$\begin{aligned}\partial\Omega_t \cup \partial\Omega_u &= \partial\Omega, & \partial\Omega_t \cap \partial\Omega_u &= \emptyset \\ \partial\Omega_q \cup \partial\Omega_T &= \partial\Omega, & \partial\Omega_q \cap \partial\Omega_T &= \emptyset\end{aligned}\quad (1)$$

This paper proposes a fracture variational approach as the basis for the current phase field model and formulation. It states that finding the saddle point is necessary to address the fracture of quasi-brittle materials while considering inertial effects [38]:

$$S[\mathbf{u}, \dot{\mathbf{u}}, T] := \int_0^{t_a} L[\mathbf{u}, \dot{\mathbf{u}}, T] dt \quad (2)$$

where the Lagrangian L is defined as:

$$L[\mathbf{u}, \dot{\mathbf{u}}, T] := \int_{\Omega/\Gamma_c} \left\{ \frac{1}{2} \rho \dot{\mathbf{u}} \cdot \dot{\mathbf{u}} - \psi_0(\boldsymbol{\varepsilon}^e(\mathbf{u}, T)) + \rho \mathbf{b} \cdot \mathbf{u} \right\} dV + \int_{\partial\Omega_t} \bar{\mathbf{t}} \cdot \mathbf{u} d\Gamma - g_c |\Gamma| \quad (3)$$

where ρ is the material density, g_c is the critical energy release rate. $\mathbf{b}$ is the body force per mass. ψ_0 is the elastic storage strain energy density which relies on the elastic strain tensor:

$$\boldsymbol{\varepsilon}^e = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^T \quad (4)$$

where $\boldsymbol{\varepsilon}$ is the total strain and $\boldsymbol{\varepsilon}^T$ is thermal expansion strain tensor. These two strains are defined as follows:

$$\boldsymbol{\varepsilon} := \frac{1}{2} (\nabla \mathbf{u} + \nabla \mathbf{u}^T) \quad \boldsymbol{\varepsilon}^T := \alpha_T \Delta T \mathbf{I} \quad (5)$$

where α_T is the coefficient of thermal expansion (CTE) of materials, and $\mathbf{I}$ is the identity matrix.

2.2. Geometric regularization and phase-field description of cracks

In the phase-field method, the concept of damage variable $d: \Omega \rightarrow [0, 1]$ is proposed to describe the extent of the material failure. The material points with ' $d = 0$ ' and ' $d = 1$ ' refer to unbroken and fully fractured states, correspondingly. Therefore, the crack surface is dispersive, and this is evident in Fig. 1. As per the image segmentation theory, the crack surface's energy contribution is determined using the volume approximation method:

$$\int_{\Gamma_c} g_c dA \approx \int_{\Omega/\Gamma_c} g_c \gamma(d; \nabla d) dV \quad (6)$$

according to the unified phase-field damage theory [37,39,40], the crack area density $\gamma(d; \nabla d)$ can be expressed as a function of the phase field variable d and its gradient ∇d .

$$\gamma(d; \nabla d) = \frac{1}{c_a} \left[\frac{1}{l_c} \alpha(d) + l_c |\nabla d|^2 \right] \quad \text{with} \quad c_a = 4 \int_0^1 \sqrt{\alpha(\beta)} d\beta \quad (7)$$

where $\alpha(d)$ is the crack geometry function. Its specific form depends on the model and will be given in Table 1. Then the regularized variational formulation is as follows: find $(\mathbf{u}, T, d)$ that minimizes the functional $S_l[\mathbf{u}, \dot{\mathbf{u}}, T] := \int_0^{t_a} \mathcal{L}_l[\mathbf{u}, \dot{\mathbf{u}}, T] dt$ with respect to T and d , subject to appropriate boundary conditions and initial conditions.

$$\begin{aligned}\mathcal{L}_l[\mathbf{u}, \dot{\mathbf{u}}, T] &:= \int_{\Omega} \left\{ \frac{1}{2} \rho \dot{\mathbf{u}} \cdot \dot{\mathbf{u}} - \psi(\boldsymbol{\varepsilon}^e(\mathbf{u}, T), d) + \rho \mathbf{b} \cdot \mathbf{u} \right\} dV \\ &+ \int_{\partial\Omega_t} \bar{\mathbf{t}} \cdot \mathbf{u} d\Gamma - g_c \int_{\Omega/\Gamma_c} \left[\frac{1}{l_c} \alpha(d) + l_c |\nabla d|^2 \right] dV\end{aligned}\quad (8)$$

The strong form differential equation can be derived from the weak form integral equation using the variational principle in the

Table 1

The crack geometry function $\alpha(d)$, the energy degradation function $\omega(d)$ and c_a of the three phase field models.

Model	AT1	AT2	PF-CZM
$\alpha(d)$	d	d^2	$2d - d^2$
$\omega(d)$	$(1 - d)^2$	$(1 - d)^2$	$\frac{(1 - d)^2}{(1 - d)^2 + a_1 d(1 - 0.5d)}$
c_a	$8/3$	2	π

following manner:

$$\begin{aligned} \operatorname{div} \cdot \boldsymbol{\sigma} + \rho \mathbf{b} &= \rho \ddot{\mathbf{u}} & \text{in } \Omega \times [0, t_f] \\ -\frac{\partial \psi}{\partial d} - g_c \frac{1}{c_a} \left[\frac{1}{l_c} \alpha'(d) - 2l_c \Delta d \right] &= 0 & \text{in } \Omega \times [0, t_f] \end{aligned} \quad (9)$$

where $\boldsymbol{\sigma} = \partial \psi / \partial \boldsymbol{\varepsilon}^e$ is the Cauchy stress. The boundary condition for displacement is as follows (the Dirichlet (or first-type) boundary condition):

$$\mathbf{u} = \bar{\mathbf{u}}, \quad \text{on } \partial \Omega_u \times [0, t_f] \quad (10)$$

The Neumann (or second-type) boundary condition is as follows for stress and phase fields.

$$\begin{aligned} \boldsymbol{\sigma} \mathbf{n} &= \bar{\mathbf{t}} & \text{on } \partial \Omega_t \times [0, t_f] \\ \frac{\partial d}{\partial \mathbf{n}} &= 0 & \text{on } \partial \Omega \times [0, t_f] \end{aligned} \quad (11)$$

The heat convection diffusion process is given in the form of heat source and heat flow: $Q_c = \int_{\Omega} (\rho c \dot{T} + \nabla \cdot \mathbf{q} - r) dV$ (12).

The thermal diffusion process satisfies the equilibrium condition:

$$\rho c \dot{T} + \nabla \cdot \mathbf{q} = r \quad \text{in } \Omega \times [0, t_f] \quad (13)$$

where c is the specific heat capacity, r is the internal heat source per volume. The boundary conditions for temperature and heat flow are as follows:

$$\begin{aligned} T &= \bar{T} & \text{on } \partial \Omega_T \times [0, t_f] \\ \mathbf{q} \cdot \mathbf{n} &= \bar{q} & \text{on } \partial \Omega_q \times [0, t_f] \end{aligned} \quad (14)$$

where $\mathbf{q}$ is the heat flux, which is proportional to the temperature gradient, and the thermal conductivity k will also be reduced due to damage evolution, to ensure that the heat flux does not propagate perpendicular to the crack surface:

$$\mathbf{q} = -k \nabla T, \quad k = (1 - d)^2 k_0 \quad (15)$$

where k_0 is the thermal conductivity of intact material.

The convective heat transfer interface on the surface plays a pivotal role in the quenching process. The convective heat transfer is directly proportional to the temperature difference between the surface and the environment:

$$\mathbf{q} \cdot \mathbf{n} = -k \nabla T \cdot \mathbf{n} = \bar{h}(T_s - T_{\infty}) \quad \text{on } \partial \Omega_q \times [0, t_f] \quad (16)$$

where $\bar{h}$ is the surface convective heat transfer coefficient, T_s is the surface temperature, and T_{∞} is the ambient temperature.

2.3. Phase field driving force and governing equations

In light of the thermal shock fracturing process of quasi-brittle materials, it is imperative that the free energy density function $\psi(\boldsymbol{\varepsilon}^e(\mathbf{u}, T, d))$ takes into consideration the decline of the energy degradation function $\omega(d)$. The subsequent equation is as follows:

$$\psi(\boldsymbol{\varepsilon}, T, d) = \omega(d) \psi_0(\boldsymbol{\varepsilon}, T) \quad (17)$$

where $\psi_0(\boldsymbol{\varepsilon}^e(\mathbf{u}, T))$ is the isotropic degenerate undamaged strain energy density function:

$$\psi_0(\boldsymbol{\varepsilon}, T) = \frac{1}{2} (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^T) : \mathbb{E}_0 : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^T) = \frac{1}{2} \bar{\boldsymbol{\sigma}} : \mathbb{C}_0 : \bar{\boldsymbol{\sigma}} \quad (18)$$

where, $\mathbb{E}_0$ and $\mathbb{C}_0$ are the material's fourth-order elastic stiffness and flexibility tensor. $\bar{\boldsymbol{\sigma}}$ is the effective stress tensor, given by the following linear elasticity relation:

$$\bar{\boldsymbol{\sigma}} = \mathbb{E}_0 : \boldsymbol{\varepsilon}^e \quad (19)$$

The energy release rate of the fracture phase field is defined as:

$$Y := -\frac{\partial \psi}{\partial d} = -\omega'(d) \mathcal{H} \quad (20)$$

where, $\mathcal{H} := \partial \psi / \partial \omega$ is the effective energy release rate. It is also regarded as the driving force for the evolution of the phase field. $\omega'(d) := \partial \omega / \partial d$ is the derivative of the energy degradation function.

Taking into account the varying mechanical responses to tensile and compressive stress conditions, the uniaxial strength under

tensile forces was used as a benchmark, leading to a redefinition of the effective release rate of energy $\mathcal{H}$ as:

$$\mathcal{H} = \frac{1}{2E_0} \langle \bar{\sigma}_1 \rangle_+^2 \quad (21)$$

where, $\bar{\sigma}_1$ is the maximum principal stress of the effective stress tensor $\bar{\sigma}$. $\langle x \rangle_\pm$ is Macaulay brackets: $\langle x \rangle_\pm := (x \pm |x|)/2$ for any $x \in \mathbb{R}$. Then the governing equation of the phase field in Eq. (9) can be written as:

$$-\omega'(d)\mathcal{H} - g_c \frac{1}{c_a} \left[\frac{1}{l_c} \alpha'(d) - 2l_c \Delta d \right] = 0 \quad (22)$$

The explicit time integration algorithm has good robustness and is particularly suitable for transient shock problems. In order to use the explicit time integration algorithm, Eq. (22) can be replaced by adding a time-dependent term (also known as viscous term) according to Ref. [41,42]:

$$\dot{d} = \begin{cases} \frac{1}{\eta} \left\langle -\omega'(d)\mathcal{H} - g_c \frac{1}{c_a} \left[\frac{1}{l_c} \alpha'(d) - 2l_c \Delta d \right] \right\rangle_+, & d < 1 \\ 0, & \text{otherwise} \end{cases} \quad (23)$$

where η is the viscous parameter that governs the evolution of the phase field. The viscous term dissipates some of the system energy. It should be noted that η needs to be small enough to make the viscous dissipation small enough.

To prevent cracks from healing, the driving force $\mathcal{H}$ needs to fulfill the following irreversible evolution condition, which ensures that d does not decrease with time:

$$\mathcal{H} = \begin{cases} \frac{1}{2E_0} \langle \bar{\sigma}_1 \rangle_+^2, & \text{if } \frac{1}{2E_0} \langle \bar{\sigma}_1 \rangle_+^2 > \mathcal{H}_n \\ \mathcal{H}_n, & \text{otherwise} \end{cases} \quad (24)$$

To better address the problem of crack initiation and quasi-brittle damage, the following expressions for the crack geometry function $\alpha(d)$, the energy degradation function $\omega(d)$ and c_a of three phase field models are considered according to the literature [39], as shown in Table 1. in which

$$a_1 = \frac{4}{\pi} \frac{l_{ch}}{l_c}, \quad l_{ch} = E_0 g_c / f_t^2 \quad (25)$$

where l_{ch} represents the Griffith or Irwin length and serves to describe the brittleness of a solid material. A smaller l_{ch} value indicates a more brittle material.

The preceding model is a phase-field cohesive zone model that takes into account inertial effects and thermodynamic coupling when dealing with complex failures and fractures of quasi-brittle materials under thermal shock loading. The aforementioned PF-CZM model is put into effect by developing the user subroutines VUMAT and VHETVAL and merging them with other subroutines, including the user field variables subroutine VUSDFLD, in the commercial FEM package Abaqus/Explicit. For further information regarding the implementation of FEM, please refer to our previously published articles [36,38].

3. Validation of the effectiveness of the phase field model

In this section, we investigate the fracture process of a single component using a phase field model under cooling conditions to

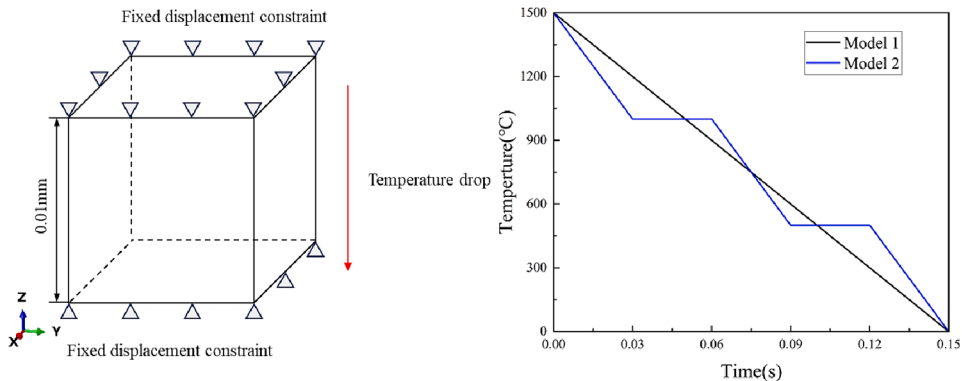


Fig. 2. Schematic of boundary conditions of the cell model and curve of temperature drop over time.

assess its validity and select the appropriate type for subsequent use. The element is constrained by a fixed displacement in the Z direction and undergoes an initial temperature of 1500 °C. Two methods of lowering the temperature were used, Model 1 was a gradual decrease over time and Model 2 was a gradual decrease in temperature. The boundary condition diagram and the temperature drop curve are shown in Fig. 2. The parameters for material and phase field utilized in simulating the cooling process of the element, through the application of the phase field technique, are presented in Table 2.

The model above is simulated using the PF-CZM, AT1 and AT2 models. The model element type is C3D8RT. The time span of simulation is 150 ms and the stable time increment is 1.7 μ s. For display time integrals, the central difference method is used for displacement and the forward difference method is used for temperature and field. The numerical solution curves were obtained for the variation of the phase field variable d and Cauchy stress σ with temperature drop as the independent variable are obtained. According to the above conditions, strict formula calculation is carried out, and the analytical solutions of phase field variable d and Cauchy stress σ are obtained. Please refer to Appendix A for further details.

Numerical and analytical solutions for the phase field variable d and the Cauchy stress σ for the three models are plotted and compared. As can be seen from the comparison in Fig. 3 (a) (b), there is a plateau segment with the temperature drop for the cooling mode 2 case when time is used as the independent variable. And the slope of the curve obtained is larger due to more temperature drop at the same time for mode 2. However, when temperature drop is used as the independent variable, it is found that the curves obtained from the two cooling modes basically overlap. Therefore, it can be concluded that the variation of the phase field variable d is independent of the cooling path for the same temperature drop. Moreover, the numerical solution of the phase field variable d is well fitted to the analytical solution, indicating the reasonableness of the above analytical solution. In addition, it is also found that the growth of the phase field variable d is nonlinear with certain thresholds for the AT1 model and the PF-CZM model. From the comparison of Fig. 4 (a) (b), it can also be concluded that the variation of the Cauchy stress σ at the same temperature drop is also independent of the cooling path. The stress peaks of the three models are different. The stress peaks of the PF-CZM model are given directly, while the other two models are affected by l_c and other parameters.

Since it is more practical to have a certain threshold for the change of phase field variable d , the energy of AT1 and CZM models will be further discussed. The distributions of kinetic energy, strain energy and fracture energy of the two models are shown in Fig. 5 and Fig. 6. The single-element cooling model has a displacement constraint in the Z direction, resulting in negligible kinetic energy. As temperature decreases, the model contracts, causing strain and an increase in strain energy. The model then generates tensile stress that reaches its own strength threshold, leading to the evolution of the phase field. The system experiences an increase in fracture energy, which corresponds to the phase field variable d . The fracture energy eventually becomes constant, resulting in the complete destruction of the model. The AT1 model energy levelled off relatively quickly. Thus only the first plateau segment is better represented when time is the independent variable. The CZM model, on the other hand, embodies both platform segments. Again, it can be shown that the variation of the energy within both models is also independent of the path of cooling and only varies with the amount of cooling.

Through the comparison of the numerical and analytical solutions of phase field variable d and Cauchy stress σ and the curve of system energy change, it can be verified that the coupled three-dimensional phase field-cohesive zone model that accounts for thermoelastic fracture established in this paper is effective. Moreover, we have opted for the PF-CZM model, which is suitable for the properties of brittle and quasi-brittle materials and its stress peak is independent of l_c . We will employ this phase field model for the following numerical analysis.

In the following, the influence of the mesh size h and the phase field scale parameter l_c on the simulated crack bandwidth is discussed. The same models, sizes, boundary conditions, cooling curves of model 1, and mesh types described in this section are used. The difference is to divide the model into meshes of different sizes and numbers. The simulation results under different mesh sizes and phase field scale parameters are shown in Fig. 7. The crack bandwidth increases with the increase of mesh size h and phase field scale parameter l_c .

4. Three-dimensional thermal shock crack formation in ceramic balls

4.1. FEM model

In this section, we consider the typical process of the aforementioned ceramic ball-in-water experiment, which is shown schematically in Fig. 8. The ceramic ball is heated to a certain temperature and then rapidly enters the cold water with a defined velocity of $V = 0.4$ m/s. The radius R of a ceramic ball is an important collective parameter that influences its quenching process. Here, the four radius dimensions with which cracks appeared in the experiments are considered: $R = 2.10$ mm, 1.00 mm, 0.56 mm, and 0.35 mm. The initial temperature of the ceramic ball is T_s and the temperature of the cold water is kept constant at T_0 . The temperature difference between the two is $\Delta T = T_s - T_0$. In this section, five different temperature differences are considered as follows, $\Delta T = -1280$ K, -780 K, -580 K, -380 K, and -280 K, in agreement with the experimental conditions of Shao et al. [11]. Typical material properties and

Table 2

The material properties and numerical parameters of the phase field modeling for ceramic ball quenching.

Parameter	E	ν	g_c	f_t	ρ	c	k_0	α
Value	370	0.22	0.0425	180	3720	880	12	8.0e-6
Unit	GPa	—	N/mm	MPa	kg/m ³	J/kg·K	W/m·K	1/K

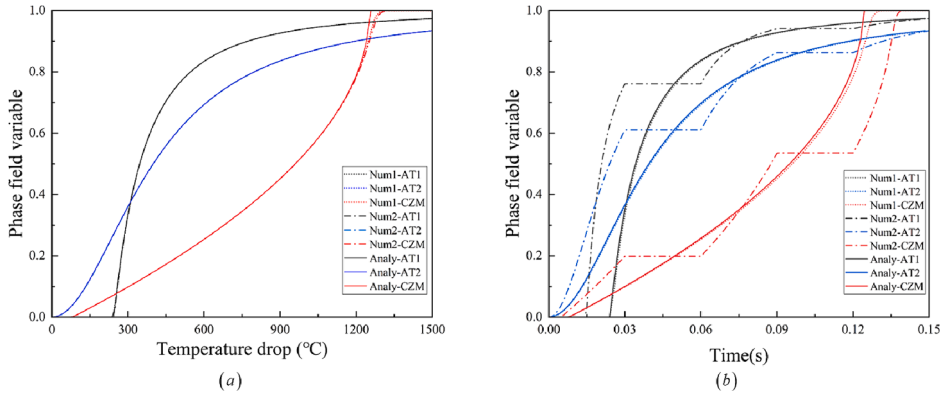


Fig. 3. Analytical and numerical solution curves of phase field variable d with temperature drop and time (a) Temperature drop is the independent variable. (b) Time is the independent variable.

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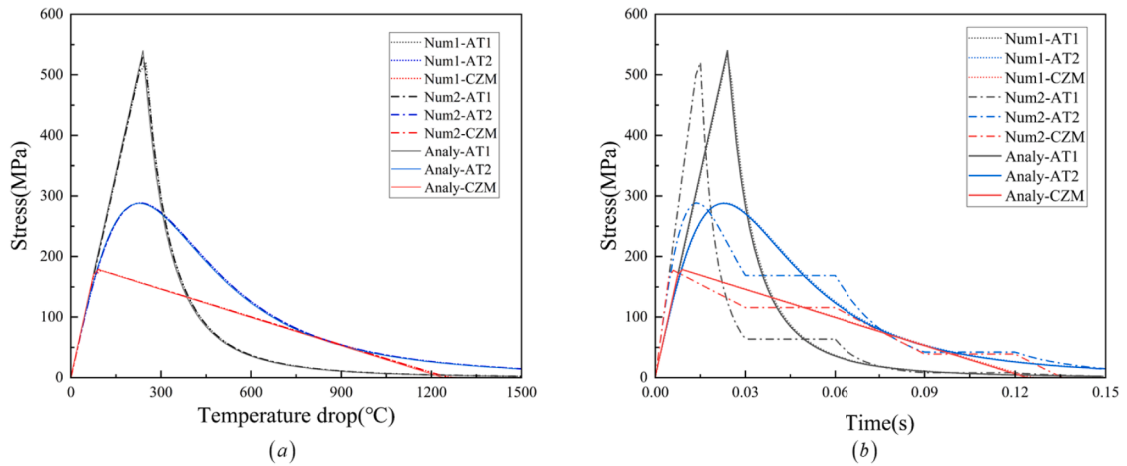


Fig. 4. Analytical and numerical solution curves of the variation of Cauchy stress σ with temperature drop and time (a) Temperature drop is the independent variable. (b) Time is the independent variable.

numerical parameters for the phase field simulation of the ceramic ball quenching are given in Table 2. In addition, the convective heat transfer coefficient between the surface of the ceramic ball and water is set to $h = 80 \text{ N}/(\text{mm}\cdot\text{s}\cdot\text{K})$. The element type in the model is C3D8RT. The element size is 0.01 mm, and the number of elements and nodes under different radii are shown in Table 3. For display time integrals, the central difference method is used for displacement and the forward difference method is used for temperature and field.

4.2. Phase field modeling results of typical quenching cracks

The results of a phase field simulation and the corresponding experimental of a typical ceramic ball ($R = 1.00 \text{ mm}$) during quenching in water ($\Delta T = -780 \text{ K}$) are given in Fig. 9. Four typical moments ($t = 3 \text{ ms}$, 5 ms , 10 ms , 20 ms) are extracted for crack extension. The first row of the figure shows the cracked phase field distribution on the surface of the ceramic ball and the crack distribution in the experiment, the second row shows a cross-sectional view of the cracked phase field distribution, and the third row shows the topology of the cracked surface in three dimensions (the region with $d > 0.95$). As can be seen from the figures, the phase field model can simulate the crack initiation and expansion process of ceramics under thermal shock very well. Due to the directional nature of the incoming water, the cracks first arise from the side of the incoming water and rapidly extend over the entire surface of the ceramic ball. At the same time, as the temperature propagates towards the center of the ceramic ball, the crack also gradually extends and expands internally, eventually forming a complex three-dimensional crack network ($t = 20 \text{ ms}$). And the results obtained from simulation are in good agreement with the experimental results.

The temperature distribution of the ceramic balls during water entry and at different moments ($t = 1 \text{ ms}$, 3 ms , 5 ms , 10 ms , 20 ms)

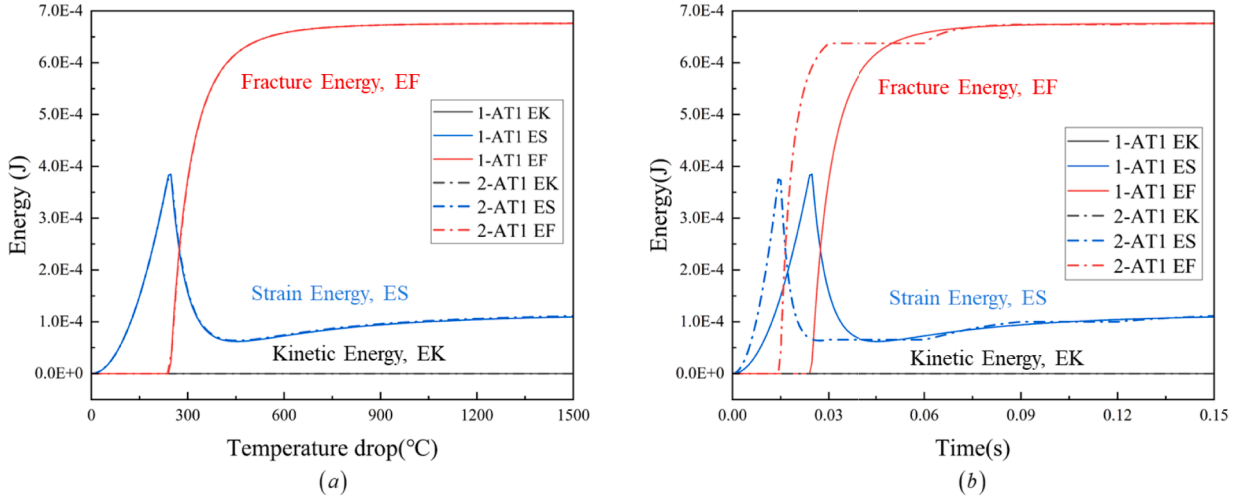


Fig. 5. The kinetic energy, strain energy and fracture energy change curves of AT1 model with temperature drop and time (a) Temperature drop is the independent variable. (b) Time is the independent variable.

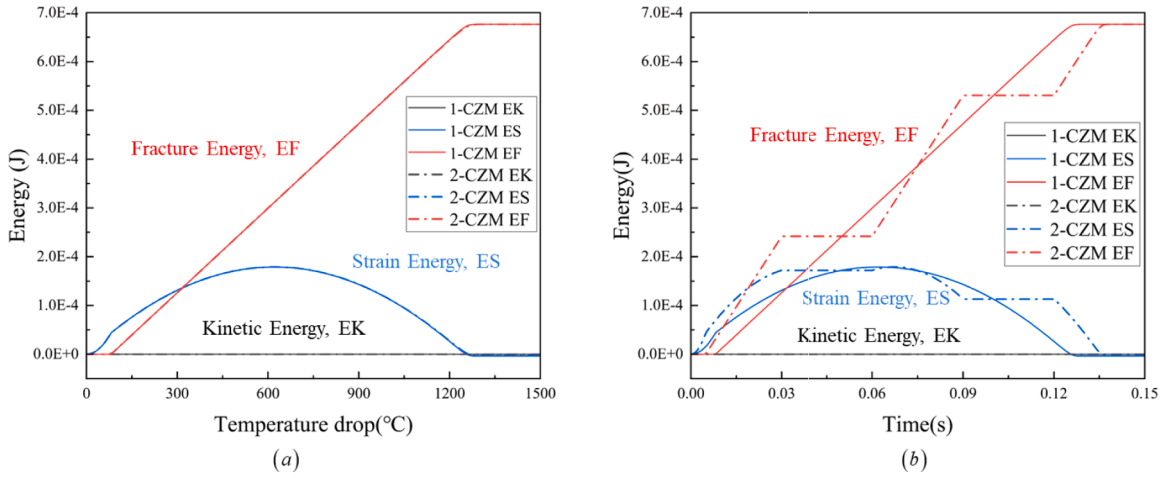


Fig. 6. The kinetic energy, strain energy and fracture energy change curves of CZM model with temperature drop and time (a) Temperature drop is the independent variable. (b) Time is the independent variable.

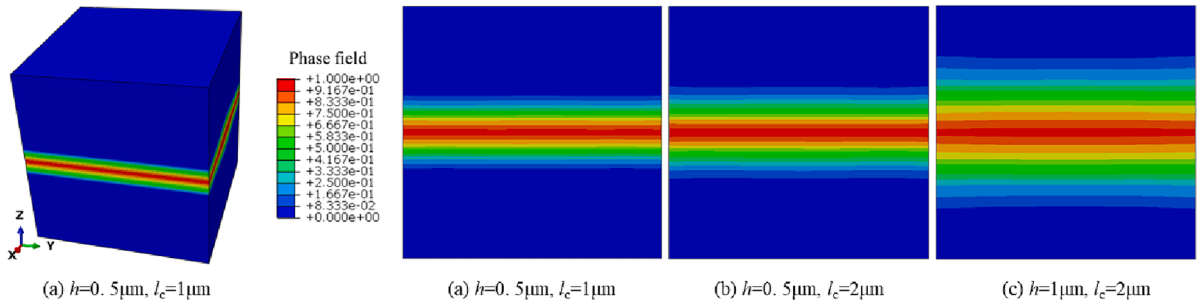


Fig. 7. The distribution of phase field values under different mesh size and phase field scale parameters. (a) $h=0.5 \mu\text{m}$, $l_c=1 \mu\text{m}$; (b) $h=0.5 \mu\text{m}$, $l_c=2 \mu\text{m}$; (c) $h=1 \mu\text{m}$, $l_c=2 \mu\text{m}$.

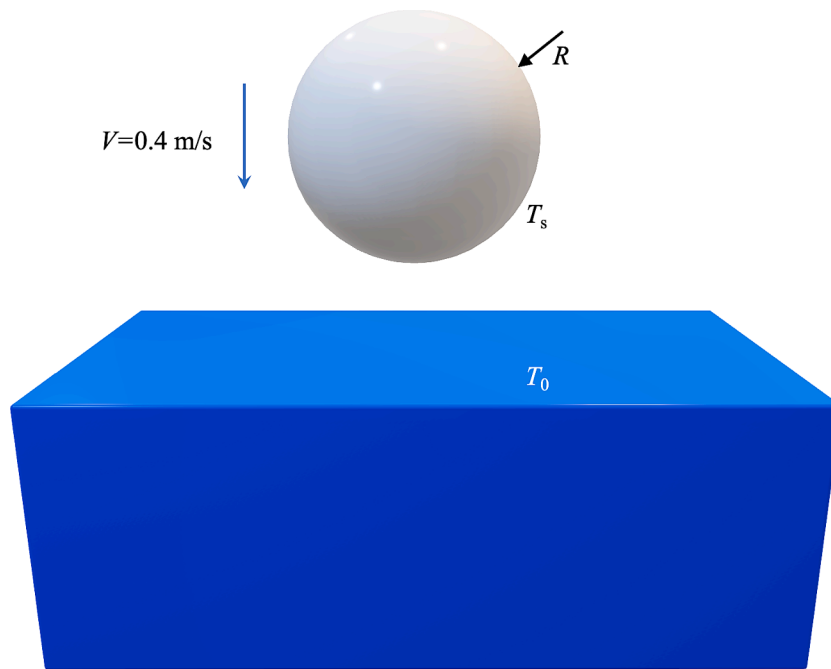


Fig. 8. Schematic diagram of a typical process of a ceramic ball entering the water with a constant velocity V .

Table 3

The number of elements and nodes under different radii.

Radius (mm)	The number of elements	The number of nodes
0.35	66,096	69,582
0.56	247,744	256,078
1	1,400,000	1,426,281
2.1	3,900,400	3,952,034

is shown schematically in Fig. 10. The region of $d > 0.95$ is removed in the figure to show the cracks. The blue color in the first row indicates below the water surface and the gray color indicates above the water surface. By $t = 5$ ms, the ceramic ball is completely in the water. The temperature propagation is consistent with the water entry process and slightly lags behind the water entry process. Due to the up-and-down asymmetry of the water entry process, the temperature distribution is also asymmetric, with the cooling region gradually propagating from the bottom to the top and from the outside to the inside. By $t = 20$ ms, the asymmetry of the temperature distribution is no longer obvious because the ceramic ball has been in the water for some time. The ceramic ball as a whole shows the cooling from the outside to the inside. The crack expansion and the temperature propagation process also remain the same, and the crack expansion area is basically in the area with a larger temperature gradient. In addition, it should be pointed out that since the effect of fracture on the heat conduction process is considered in this paper, the phase field variables discount the heat conduction coefficient, so that at the late stage of crack expansion ($t = 20$ ms), a clear discontinuity in temperature on both sides of the crack can be observed, which is consistent with our intuitive understanding.

The distribution of the mises maximum principal stress in the ceramic ball at four typical moments ($t = 3$ ms, 5 ms, 10 ms, 20 ms) is given in Fig. 11. The ceramic ball is dissected and kept in a 1/4 area for more visual observation of the cracks and stress distribution inside the ceramic ball. At the early stage of quenching ($t = 3$ ms), most of the area is not yet cracked or only very short cracks sprout. Due to the cooling shrinkage, the ceramic ball surface as a whole has a high tensile stress, and under this tensile stress, a large number of initial cracks are generated. As time progresses, the crack expands from the ceramic surface to the interior, the surface tensile stress is released, and the high-stress area is concentrated near the crack tip and dominates the further expansion of the crack. Similar cracks compete directly at this point, with long cracks shielding nearby short cracks, and only a few “dominant” cracks eventually expand farther.

The crack distribution on the surface of the ceramic balls obtained from the numerical simulation of the phase field for different ceramic ball radii and different quenching temperatures are given in Fig. 12, as well as a comparison with the experimental results. As can be seen from the table, the crack distribution densities obtained from the phase field simulation are in good agreement with the experimental results for all working conditions, which verifies the reliability of the phase field simulation. The same conclusion can be drawn from the experiments and simulations: the ceramic ball quenching process has a critical quenching temperature difference and a critical ceramic ball size, and the two are mutually influenced, when the quenching temperature difference or the ceramic ball radius is

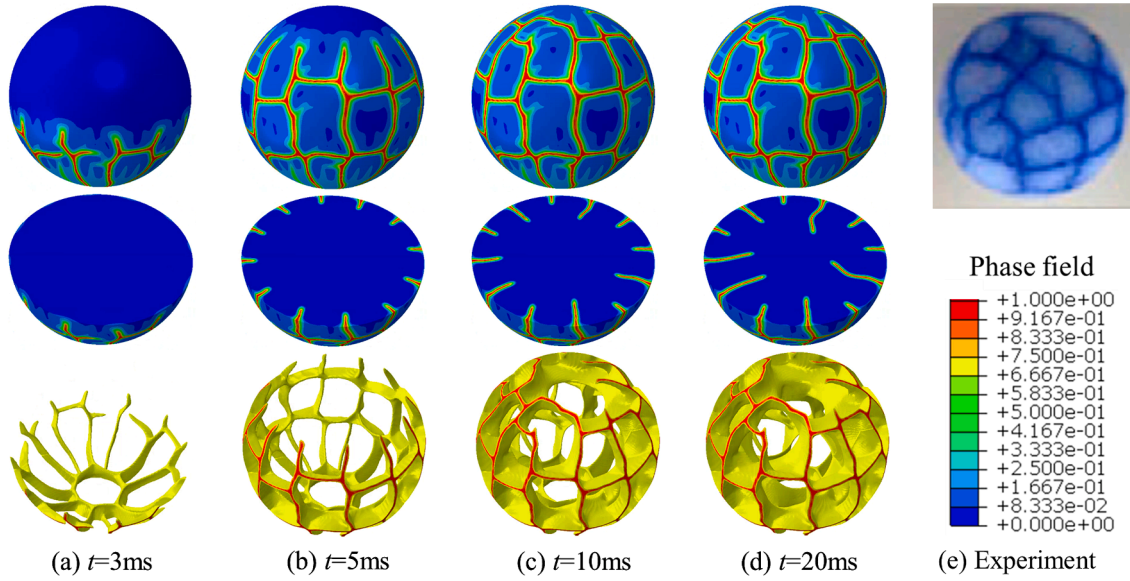


Fig. 9. Thermal shock crack propagation in a ceramic ball ($R = 1.0$ mm) under quenching conditions ($\Delta T = -380$ K) in water at several typical moments: (a) $t = 3$ ms; (b) $t = 5$ ms; (c) $t = 10$ ms; (d) $t = 20$ ms; and (e) Experiment. The first row shows the crack phase-field distribution on the surface of the ceramic ball and the crack distribution in the experiment, the second row shows a cross-sectional view of the crack phase-field distribution, and the third row shows the topology of the three-dimensional spatial crack surface (the region with $d > 0.95$).

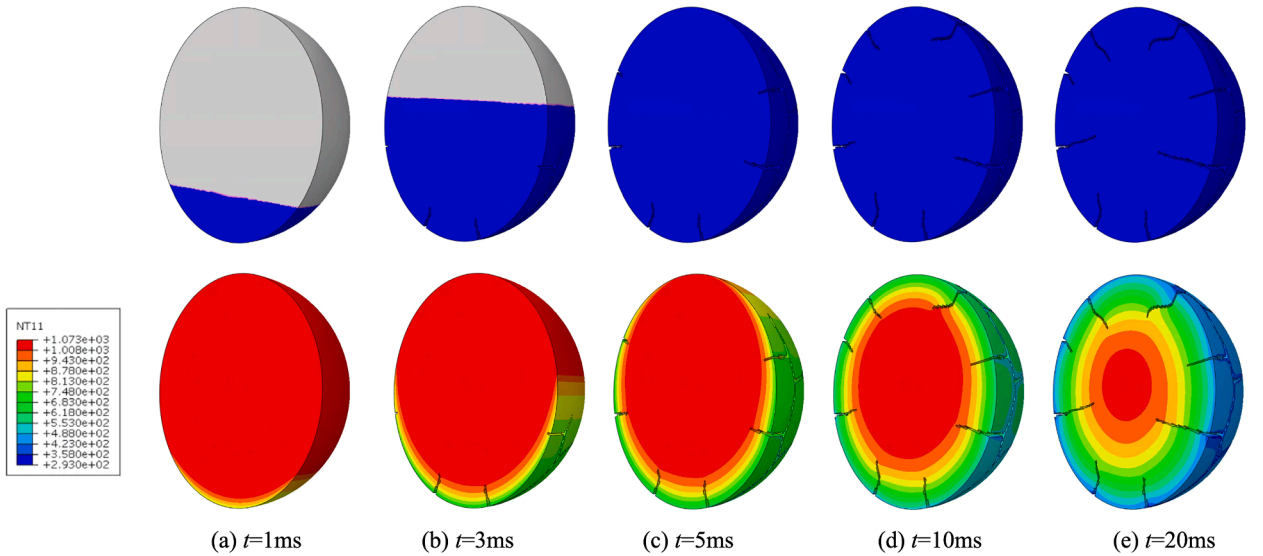


Fig. 10. Schematic diagram of the water entry process of the ceramic ball and the temperature distribution at different moments, with the region of $d > 0.95$ removed to show the cracks: (a) $t = 1$ ms; (b) $t = 3$ ms; (c) $t = 5$ ms; (d) $t = 10$ ms; and (e) $t = 20$ ms. The blue color in the first row indicates below the water surface and the gray color is above the water surface.

small to a certain degree, the ceramic ball surface will no longer produce cracks.

4.3. Statistical analysis

Firstly, the influence of the randomness of the mesh is discussed. Consider the case where the radius of the ceramic ball is 1 mm and the temperature drops to 1280 K. Fig. 13 shows the simulation results of surface cracks on ceramic balls and the statistical results of the number of crack blocks by three different meshing methods with the same mesh size. Although there are differences in the specific trend of cracks under the three mesh partition methods, the number of crack blocks on the whole surface of the ceramic ball is very close. The randomness of the mesh will affect the specific direction of the cracks, but it will not have a great impact on the simulation

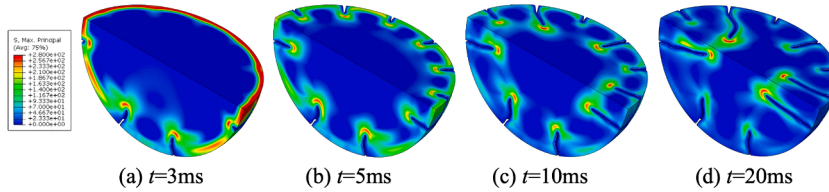


Fig. 11. The distribution of the maximum principal stresses in the ceramic ball at four typical moments: (a) $t = 3$ ms; (b) $t = 5$ ms; (c) $t = 10$ ms; and (d) $t = 20$ ms. The ceramic ball is dissected and kept in a 1/4 area for more visual observation of the cracks and stress distribution inside the ceramic ball.

results.

In order to quantitatively study the influence law of ceramic ball size and quenching temperature on the distribution of cracks on the surface of ceramic balls, the number of blocks cracked on the surface of ceramic balls was counted for each working condition, as shown in Fig. 14. The bars with oblique stripes inside are all phase field simulation results, and those without stripes are experimental results. It can be seen that the simulation results are in good agreement with the experimental results, indicating that the phase field simulation has a good prediction ability for the quenching fracture of ceramic balls. In addition, the number of cracks and the number of splintered surface blocks increase significantly with the increase in quenching temperature. For simplicity, we do not consider the effect of material parameters with temperature. This will affect the specific number of cracks, but has little effect on the conclusion of the overall regularity. This is the deficiency of our work. We will complement our work by referring to the work of many scholars [43–45] on the influence of temperature on material parameters.

Furthermore, we computed the mean surface area of the cracking blocks created by thermal shock at various temperatures and radii of ceramic balls. This computation was determined by dividing the total surface area of the ceramic ball by the number of cracking blocks:

$$\bar{S} = \frac{S_{\text{ball}}}{N_{\text{blocks}}} = \frac{4\pi R_{\text{ball}}^2}{N_{\text{blocks}}} \quad (26)$$

Fig. 15 shows the variation curves of the average surface cracked block area with quenching temperature for different working conditions in experiments and simulations. It can be seen that from a statistical perspective, under the same temperature difference, thermal shock causes the formation of similar-sized crack blocks on the surface of ceramic balls. The average area of the cracked block is independent of the size of the ceramic ball. In addition, under logarithmic coordinates, the average area of the cracked block is directly proportional to the temperature difference during quenching, namely:

$$\lg(\bar{S}) = -1.88\lg(\Delta T) + 5.02 \quad (27)$$

where, the unit of $\bar{S}$ is mm^2 , and the unit of ΔT is K.

5. Conclusions

In this paper, a coupled three-dimensional phase field-cohesive zone model considering thermoelastic fracture is developed to well reproduce the ceramic ball quenching experiments. The process of the influence of temperature and ceramic ball size on the surface crack formation of ceramic balls during quenching is investigated. The main conclusions are:

- (1) The phase field-cohesive zone model can accurately describe the three-dimensional fracture process of quasi-brittle materials under thermal shock, encompassing three-dimensional crack initiation, extension, confluence, and bifurcation.
- (2) The distribution density of cracks is defined for a given difference in quenching temperature and size of ceramic balls, although the pattern of surface cracks appears somewhat haphazard.
- (3) There exists a limiting size for the quenching of ceramic balls, whereby if the radius of the ceramic ball is smaller than this size, the formation of surface cracks in the quenching process ceases to occur.

In conclusion, the coupled phase field model developed in this paper can effectively capture the intricate three-dimensional fracture phenomena that occur during the quenching of ceramic balls. Moreover, the model exhibits good reproducibility for experimental results and has the potential to predict fracture processes in other quasi-brittle materials subjected to thermodynamically coupled loading.

CRediT authorship contribution statement

Tao Wang: Writing – original draft, Validation, Software, Methodology, Data curation. **Yichen Zhang:** Writing – original draft, Visualization. **Haoyue Han:** Writing – review & editing, Validation, Supervision. **Lei Wang:** Writing – review & editing, Visualization, Validation. **Xuan Ye:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Writing – review &

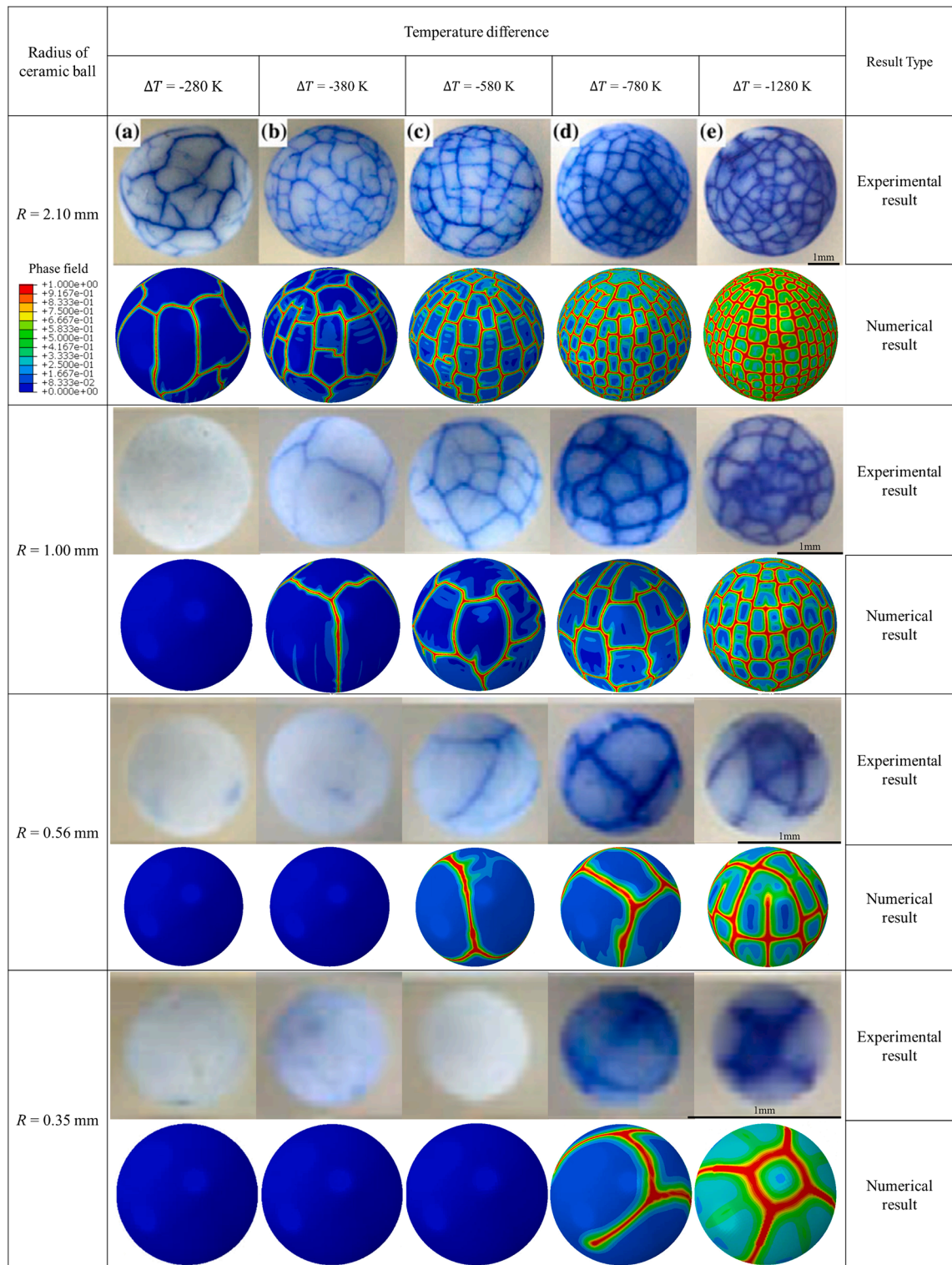


Fig. 12. Crack distribution on the surface of ceramic balls simulated by the phase field model at different quenching temperature differences ($\Delta T = -280$ K, -380 K, -580 K, -780 K and -1280 K) and different ceramic ball's radii ($R = 2.10$ mm, 1.00 mm, 0.56 mm and 0.35 mm) and its comparison with previous experimental results [11].

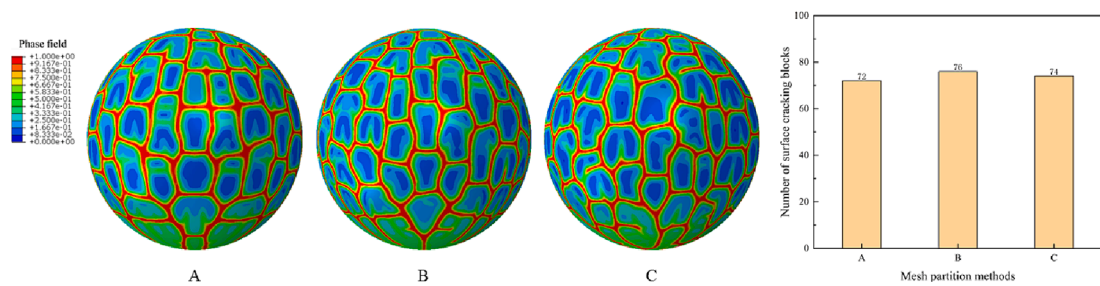


Fig. 13. The simulation results of surface cracks on ceramic balls and the statistical results of the number of crack blocks by three different meshing methods with the same mesh size.

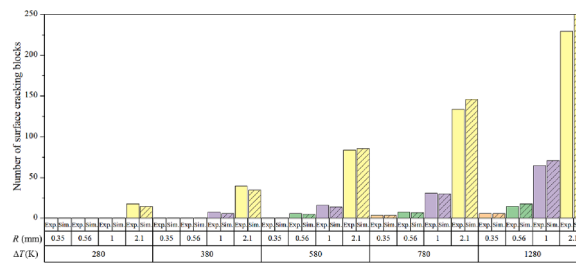


Fig. 14. The number of cracked blocks on the surface of the ceramic balls was statistically obtained for various working conditions of the experiments and simulations.

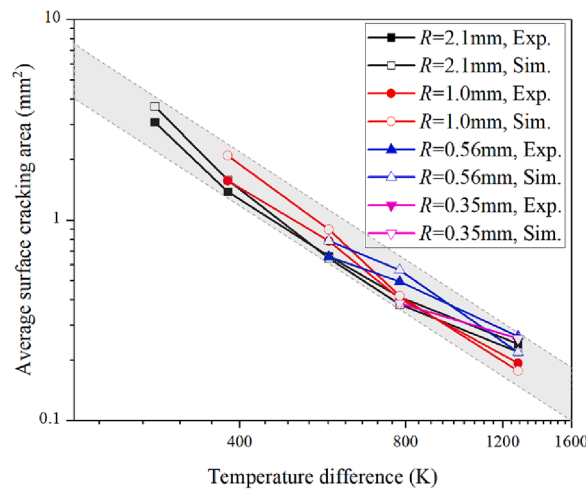


Fig. 15. Variation curves of average surface cracked block area with quenching temperature for different working conditions in experiments and simulations.

editing. **Zhuo Zhuang:** Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Analytical solution of phase field for 1D thermal expansion crack of a single element.

Consider a single square element that constrains normal degrees of freedom in one direction and the other four faces are free, as shown in Fig. 2. Given a certain temperature drop ΔT , the resulting thermal strain is:

$$\varepsilon^\theta = \alpha \Delta T \mathbf{I} \quad (\text{A1})$$

Assuming the material is linearly elastic and isotropic, and the total strain generated in two unconstrained directions is the same, denoted as ε_x , then the total strain of the element is:

$$\boldsymbol{\varepsilon} = [\varepsilon_x \quad \varepsilon_x \quad 0 \quad 0 \quad 0 \quad 0] \quad (\text{A2})$$

thus, the elastic strain of the element in the current state can be obtained:

$$\boldsymbol{\varepsilon}^e = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^\theta = [\varepsilon_x - \alpha \Delta T \quad \varepsilon_x - \alpha \Delta T \quad -\alpha \Delta T \quad 0 \quad 0 \quad 0] \quad (\text{A3})$$

according to the stress state of the unit, it can be seen that:

$$\boldsymbol{\sigma} = \mathbf{C}\boldsymbol{\varepsilon} = [0 \quad 0 \quad \sigma_z \quad 0 \quad 0 \quad 0] \quad (\text{A4})$$

where $\mathbf{C}$ is the elastic matrix of the material, determined by its Young's modulus E and Poisson's ratio ν . Expand the equation above to obtain:

$$(1 - \nu)(\varepsilon_x - \alpha \Delta T) + \nu(\varepsilon_x - \alpha \Delta T) - \nu \alpha \Delta T = 0 \quad (\text{A5})$$

$$\sigma_z = \frac{E}{(1 + \nu)(1 - 2\nu)} [2\nu(\varepsilon_x - \alpha \Delta T) - (1 - \nu)\alpha \Delta T] \quad (\text{A6})$$

thus, it can be obtained that:

$$\varepsilon_x = (1 - \nu)\alpha \Delta T \quad (\text{A7})$$

$$\sigma_z = -E\alpha \Delta T \quad (\text{A8})$$

The effective energy release rate $\mathcal{H}$ that drives the phase field evolution can be obtained:

$$\mathcal{H} = \frac{1}{2E} \langle \sigma_1 \rangle_+^2 = \frac{1}{2} E \alpha^2 (\Delta T)^2 \quad (\text{A9})$$

Ignoring the viscosity term, the element spatial gradient term is 0, which can be obtained:

$$\frac{\omega'(d)}{\alpha(d)} = -\frac{g_c}{c_\alpha l_c \mathcal{H}} \quad (\text{A10})$$

Substituting the driving force $\mathcal{H}$ into the phase field evolution equation (23), Then the phase field variable d can be solved.

For PF-CZM model:

$$\begin{cases} \left[(1-d)^2 + a_1 d \left(1 - \frac{1}{2} d \right) \right]^2 = \frac{E^2 \alpha^2 \Delta T^2}{f_t^2} \\ 0 \leq d \leq 1 \end{cases} \quad (\text{A11})$$

The above equation can be solved for the determined phase field evolution value d for a given temperature drop ΔT .

For AT1 model:

$$\begin{cases} d = 1 - \frac{g_c}{c_\alpha l_c E \alpha^2 \Delta T^2} \\ d > 0 \end{cases} \quad (\text{A12})$$

For AT2 model:

$$d = \frac{1}{1 + \frac{g_c}{l_c E \alpha^2 \Delta T^2}} \quad (13)$$

Appendix B. Python source code used to obtain the analytical solutions of 1D thermal expansion crack of a single element

The Python source code is provided to calculate the analytical solutions of the 1D thermal expansion crack of a single element, including the variation of phase field variables d and stress σ with temperature drop ΔT according to the equations in Appendix A.

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